Alex Chen

AI & Machine Learning Engineer

Email: alex.chen@example.com | Phone: +1 (555) 234-5678 | LinkedIn: linkedin.com/in/alexchen-ai | GitHub: github.com/alexchen-dev

Professional Summary

Results-driven Machine Learning Engineer with a solid foundation in Deep Learning, Natural Language Processing, and MLOps. Experienced in building Retrieval-Augmented Generation (RAG) systems and deploying scalable PyTorch models using Docker and FastAPI.

Technical Skills

Programming Languages: Python, SQL, C++, Bash
AI / Machine Learning: PyTorch, TensorFlow, Scikit-learn, Hugging Face, Transformers, NLP, LangChain, Pandas, NumPy, OpenCV
MLOps & Cloud: Docker, MLflow, AWS, Git, GitHub Actions, CI/CD
Databases & Frameworks: PostgreSQL, FastAPI, Redis

Work Experience & Internships

Machine Learning Intern – *Cognitive AI Solutions* (May 2025 – August 2025)

- Engineered and fine-tuned BERT-based NLP models for document classification, achieving 93.4% accuracy across 50,000 corporate records.
- Built high-throughput inference APIs using FastAPI and Docker, reducing average response latency by 32% (under 80ms).
- Automated model training pipelines using MLflow and GitHub Actions, cutting continuous retraining cycle time by 4 hours weekly.

Key Technical Projects

Neural RAG Knowledge Assistant | *Python, LangChain, PyTorch, FAISS, FastAPI, Docker*

- Architected an enterprise question-answering pipeline using LangChain, Hugging Face embeddings, and FAISS vector database.
- Deployed containerized microservice on AWS EC2 handling 1,500+ daily queries with sub-second retrieval times.
- Implemented semantic chunking and reranking, improving document retrieval precision by 24% over baseline BM25.

Autonomous Vision Defect Detector | *PyTorch, OpenCV, Scikit-learn, Matplotlib*

- Trained a convolutional neural network (CNN) for surface anomaly detection, achieving a 0.96 F1-score on an industrial benchmark dataset.
- Integrated OpenCV pre-processing filters and automated evaluation scripts with detailed confusion matrices.

Education

Bachelor of Technology in Computer Science (AI & ML), Apex Institute of Technology (2022 - 2026) – GPA: 3.85 / 4.0

Certifications & Achievements

- DeepLearning.AI Deep Learning Specialization (Coursera)
- AWS Certified Cloud Practitioner (Amazon Web Services)
- Kaggle Competitions Master – Top 5% in Tabular ML Hackathons